1 A 3.8 kg block on a plane inclined at 15° has initial speed of 5.0 m s^{-1} when a horizontal force *F* of magnitude 25 N acts on it as shown in Fig. 1.1.

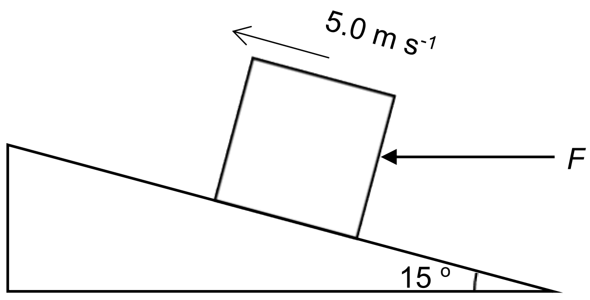

Fig. 1.1

(a) Determine the normal contact force exerted by the plane on the block.

Considering forces acting in the direction perpendicular to plane,

$$F_{net} = 0$$

$$N = F\sin\theta + mg\cos\theta$$

$$= 25\sin 15° + 3.8(9.81)\cos 15°$$

$$= 42 \text{ N}$$

normal contact force = ... N [2]

(b) The coefficient of kinetic friction, μ, is 0.40 and the kinetic friction, *f*, is related to the normal contact force, *N*, by the equation

$$f = \mu N .$$

(i) Calculate the magnitude of the acceleration of the block.

By Newton's second law of motion, taking downslope as positive,

$$F_{net} = ma$$

$$mg\sin\theta + f - F\cos\theta = ma$$

$$mg\sin\theta + \mu N - F\cos\theta = ma$$

$$a = g\sin\theta + \frac{\mu N - F\cos\theta}{m}$$

$$= 9.81\sin 15° + \frac{0.40(42) - 25\cos 15°}{3.8}$$

$$= 0.66 \text{ m s}^{-2}$$

acceleration = ... m s^{-2} [2]

(ii) Determine the distance travelled by the block before it comes to rest.

$$v^2 = u^2 + 2as$$
$$s = -\frac{u^2}{2a}$$
$$= -\frac{5.0^2}{2(-0.66)}$$
$$= 19 \text{ m}$$

distance = ... m [2]

(iii) Static friction is a force that acts on an object when it is at rest. It must be overcome before the object can start moving.

Considering that the maximum possible static friction between the object and the plane is equal to the value of the kinetic friction, *f*, state and explain, whether the block will slide down the plane after coming to rest if force *F* continues to act on the block.

component of F parallel to plane, $F_{//} = F\cos\theta$
$$= 25\cos 15°$$
$$= 24 \text{ N}$$

component of weight parallel to plane, $W_{//} = mg\sin\theta$
$$= 3.8(9.81)\sin 15°$$
$$= 9.6 \text{ N}$$

Since the component of F parallel to plane is greater than the component of weight parallel to plane, the block has tendency to slide up. However, as there is friction between the block and surface of plane and the maximum static friction, f,

$$f = \mu N$$
$$= 0.40(42)$$
$$= 17 \text{ N}$$

Hence friction acts parallel to plane down the slope and the amount of friction required, f',

$$F_{net} = 0$$
$$F_{//} = W_{//} + f'$$
$$f' = F_{//} - W_{//}$$
$$= 24 - 9.6$$
$$= 14 \text{ N} < f$$

Therefore block remains at rest with static friction of 14 N acting on it.

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.... [2]

[Total: 8]

2 (a) State an example of how the motion of an object can be undergoing the following